

Transformer-Based Prediction of Lithium-Ion Battery State of Health

Abstract

This study investigates a transformer-based neural network for predicting the state of health (SOH) of lithium-ion batteries. The model integrates a denoising autoencoder, positional encoding, and a transformer encoder, trained end-to-end on NASA-derived battery degradation data. Model performance is evaluated using relative error, mean absolute error (MAE), and root mean square error (RMSE). Results indicate that increasing hidden feature dimensionality improves predictive accuracy, with a 21-dimensional hidden layer achieving optimal performance.

Introduction

Accurate estimation of lithium-ion battery SOH is critical for energy storage reliability. Being able to model and accurately represent battery degradation is important because lithium-ion batteries are used in nearly every aspect of modern life. Gaining a better understanding of how different factors contribute to a battery's state of health can potentially enable us to extend battery life while also ensuring their safe operation. Machine learning models are founded on the perceptron, a probabilistic model introduced by Rosenblatt (1958) for mapping inputs to outputs. Recurrent neural networks (RNNs) later extended these models to handle sequential data, capturing temporal dependencies (Elman, 1990). Transformer architectures, originally developed for sequential data tasks, overcome limitations of RNNs by using self-attention to model long-range dependencies efficiently (Vaswani et al., 2017). Prior work combines denoising autoencoders with transformers for remaining useful life prediction (Zhang et al., 2022), motivating the approach in this study.

Dataset

The dataset used in this study is the *Lithium-ion battery degradation dataset* from Kaggle, which contains cycle-level measurements of voltage, current, temperature, internal resistance, and capacity mapped to state of health (SOH) values (programmer3, n.d.). The dataset, derived from the NASA Ames Prognostics Center of Excellence, contains cycle-level summaries for three batteries. One battery was used for training, a second for validation, and a third for final testing.

Methodology

The model comprises a denoising autoencoder, positional encoding, and transformer encoder, trained end-to-end with the Adam optimizer and mean squared error loss. Positional encoding was adapted from an open-source PyTorch implementation (sayef, n.d.). Hidden feature dimensionality was varied (7, 14, 21, 28) to assess its effect on predictive accuracy. Relative error, MAE, and RMSE were used to evaluate performance. Relative error measures the normalized difference between predicted and true SOH values, MAE quantifies the average magnitude of prediction errors, and RMSE penalizes larger errors more heavily.

Results and Discussion

Model evaluation on the test battery shows that increasing hidden feature dimensionality improves performance up to an optimal point. Table 1 summarizes the results.

Table 1

Model Performance Across Different Hidden Dimensions

Metric	D_Model_7	D_Model_14	D_Model_21	D_Model_28
Relative Error	0.012	0.013	0.010	0.011
Mean Absolute Error	0.060	0.067	0.050	0.057
Root Mean Square Error	0.081	0.089	0.064	0.073

The 21-dimensional model achieves the lowest relative error, MAE, and RMSE, indicating optimal predictive performance. Higher dimensionality (28) offers no further improvement, suggesting potential overfitting.

Conclusion

A transformer-based architecture with a denoising autoencoder and positional encoding effectively predicts lithium-ion battery SOH. Hidden feature dimensionality significantly affects model accuracy, with 21 features providing the best balance between complexity and generalization. Historical context from perceptrons and RNNs highlights the evolution of machine learning methods for sequential data, underscoring the transformer's advantages. This approach supports reliable SOH estimation and informs future battery prognostics research.

References

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